Logarithmic functions

1 Consider the function $y = \log_2 x$.

a Complete the table of values.

x	$\frac{1}{2}$	1	2	4	16
y					

b Sketch a graph of the function.

c State the equation of the vertical asymptote.

d State whether the following elements are key features of the graph of $y = \log_2 x$:

i A y-intercept

ii A horizontal asymptote

iii An x-intercept

iv A lower limiting value

v An upper limiting value

2 Sketch the graph of $y = \log_5 x$.

3 Consider the function $y = \log_3 x$.

a Complete the table of values.

x	$\frac{1}{243}$	$\frac{1}{3}$	1	3	9	81
y						

b Is $y = \log_3 x$ an increasing or decreasing function?

c Describe the behavior of $y = \log_4 x$ as x approaches 0.

d State the value of y when $x = 0$.

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Inverses of logarithmic functions

4 Consider the function $y = \log_2 x$.

a Complete the table of values for $y = \log_2 x$:

x	1	2	4	8
y				

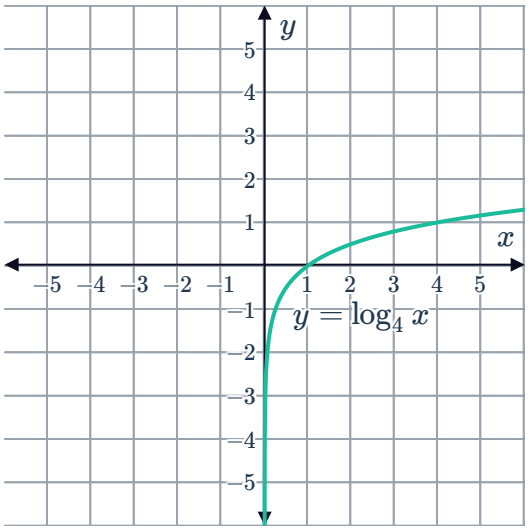
b Complete the table of values for the inverse function of $y = \log_2 x$:

x				
y				

c Sketch the graph of $y = \log_2 x$ and its inverse function on the same coordinate plane.

d State the equation of the inverse function of $y = \log_2 x$.

5 Consider the graph of $y = \log_4 x$:
Sketch the graph of the inverse of $y = \log_4 x$.



6 For each of the following functions:

i Graph $f(x)$, the line $y = x$ and the inverse of $f(x)$ on the same coordinate plane.

ii State the equation of the inverse function.

a $f(x) = 4^x$ **b** $f(x) = \left(\frac{1}{4}\right)^x$ **c** $f(x) = \left(\frac{1}{5}\right)^x$

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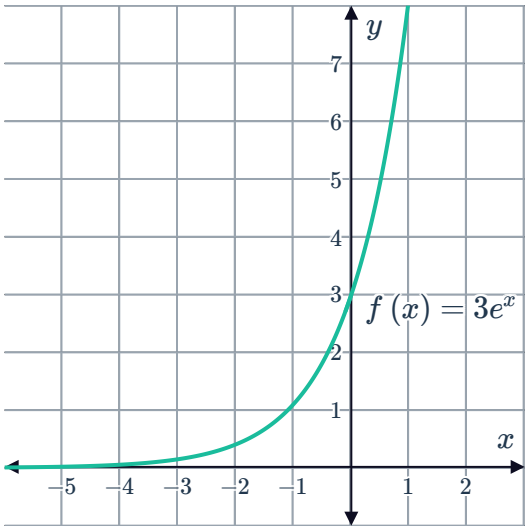
Natural logarithmic functions

7 Describe the logarithmic function known as $\ln x$ in terms of its base.

8 Consider the graph of $f(x) = 3e^x$:

a Sketch the graph of the inverse of $f(x)$.

b State the equation of the inverse function.



9 Consider the function $f(x) = 2e^x - 5$ for all $x \in \mathbb{R}$.

a State the equation of the inverse function.

b Using a graphing calculator, state the coordinates of the points at which the graphs of $y = f(x)$ and $y = f^{-1}(x)$ intersect. Round all values to three decimal places.

Transformations of logarithmic functions

10 If the function $f(x) = \log_3 x$ is translated 5 units to the right, state the equation of the resulting function.

11 Sketch the graph of $y = \log_3 x$ after it has been translated 4 units down.

12 For each of the following functions:

i State the equation of the function after it has been translated.

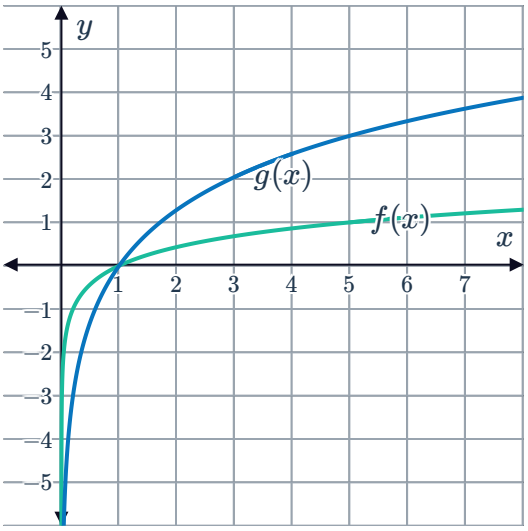
ii Sketch the translated graph.

a $y = \log_5 x$ translated downwards by 2 units.

b $y = \log_3(-x)$ translated upwards by 2 units.

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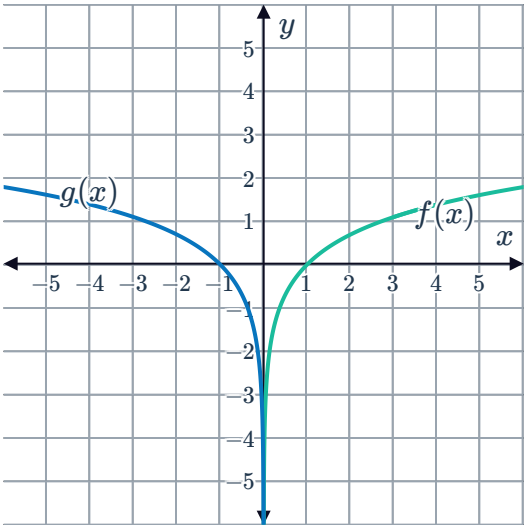
13 Consider the graphs of $f(x) = \log_5 x$ and $g(x)$:
Find the equation of $g(x)$.



14 Consider the graphs of $f(x) = \ln x$ and $g(x)$:

a Describe the transformation that has been performed on the graph of $f(x) = \ln x$ to obtain $g(x)$.

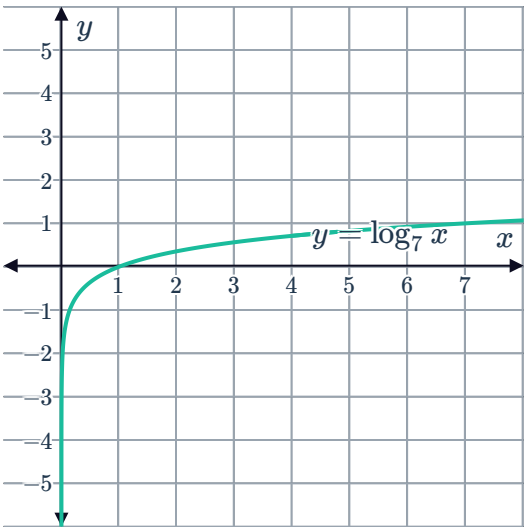
b State the equation for $g(x)$.



15 Consider the graph of $y = \log_7 x$:

a Describe the transformation needed to obtain the graph of $y = -3\log_7 x$ from $y = \log_7 x$.

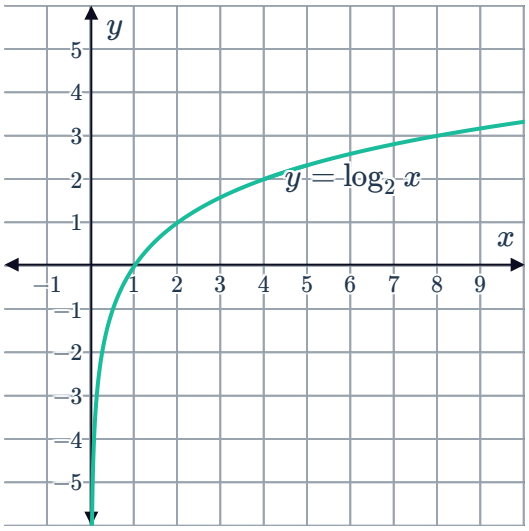
b Sketch the graph of $y = -3\log_7 x$ on the same coordinate plane as $y = \log_7 x$.



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16 Given the graph of $y = \log_2 x$, sketch the graph of the following functions:

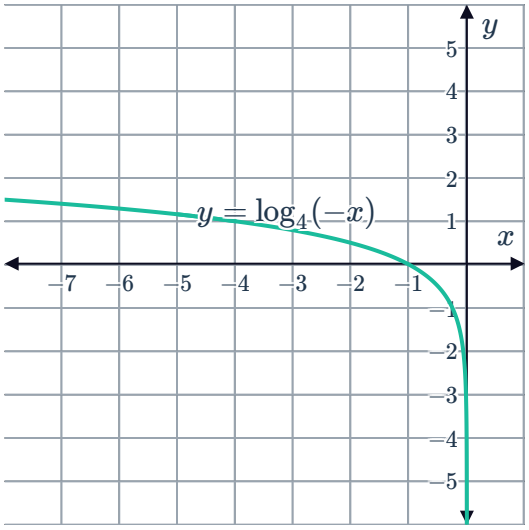
a $y = \frac{1}{3}\log_2 x$ **b** $y = -\frac{1}{2}\log_2 x$



17 Consider the graph of the function $y = \log_4(-x)$:

a Complete the table of values for $y = \log_4(-x)$:

x	-4	-1	$-\frac{1}{4}$
$\log_4(-x)$			



The graph of $y = \log_4(2 - x)$ can be obtained from the original graph by transforming it.

b Complete the table of values for $y = \log_4(2 - x)$:

x	-2	1	$\frac{7}{4}$
$\log_4(2 - x)$			

c Describe the transformation from $y = \log_4(-x)$ to $y = \log_4(2 - x)$.

18 Consider the function $f(x) = \log_3(x - 4)$.

a State the equation of the vertical asymptote of $f(x)$.

b State the coordinates of the x-intercept of the function.

c Determine the exact value of $f(7)$.

d Sketch a graph of $f(x) = \log_3(x - 4)$.

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19 Consider the function $f(x) = \log_3(2 - x)$.

a State the equation of the vertical asymptote of $f(x)$.

b State the coordinates of the x-intercept of the function.

c Determine the exact value of $f(-7)$.

d Sketch a graph of $f(x) = \log_3(2 - x)$.

20 The graph of $y = \log_4 x$ has a vertical asymptote at $x = 0$. By considering the transformations that have taken place, state the equation of the vertical asymptote of the following functions:

a $y = 2\log_4 x - 4$

b $y = 2\log_4 x$

c $y = \log_4(x - 5)$

d $y = -\log_4 x$

e $y = \log_4(x + 3) - 2$

f $y = 3\log_4 x + 2$

21 For each of the following functions:

i Solve for the x-intercept.

ii State the equation of the vertical asymptote.

iii Sketch the graph of the function.

a $f(x) = 4\log_2(x - 7)$

b $f(x) = -\log_4(x + 4)$

c $f(x) = \log_2(x - 1) - 4$

d $y = -2\log_3(x + 8)$

e $y = -\log_4(x + 9) + 1$

f $y = -4\log_3(x - 1) - 4$

g $y = \log_2(-x - 1) - 1$

h $y = -\log_2(-x + 2) + 2$

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